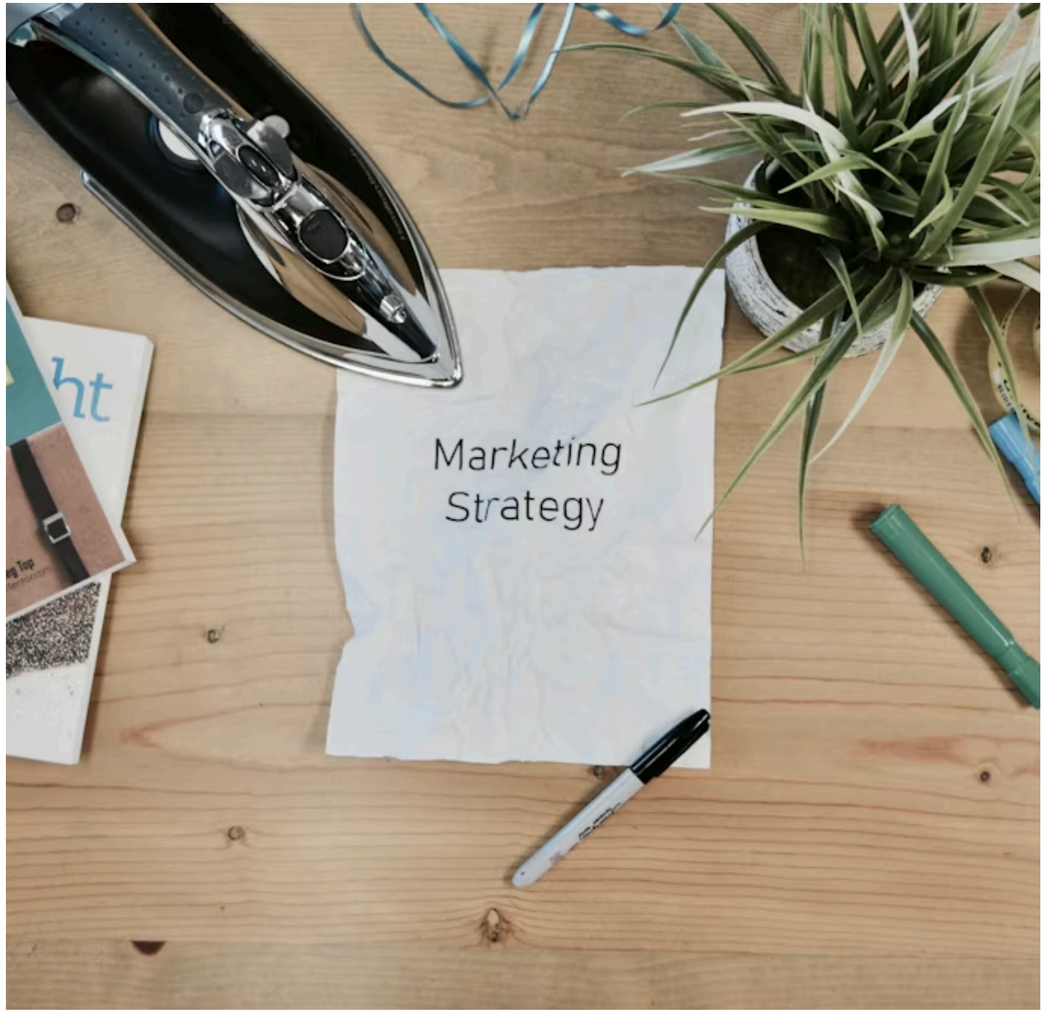


PROJECT 2

MARKETING CHANNEL PERFORMANCE ANALYSIS: BOOTCAMP CAMPAIGNS

Statistic for Business

PRESENTED BY MUNAWAR

BACKGROUND

As a Data Analyst, I explore the effectiveness of digital campaigns for Bootcamp Class sales as a key driver of company revenue growth through various advertising channels. This analysis focuses on evaluating social media budget allocations to identify the most responsive audience segments and the channels that deliver the most consistent monthly performance.

VARIABLE DESCRIPTION

CampaignID : Campaign Identity
Channel : The platform where the ads run.
AudienceType : Type of targeting audience
AdBudgetK : The advertising budget spent.
RevenueK : The income generated by generated
ConversionRate : Conversion Rate
Month : Month
CTR : Click Through Rate

KEY OBJECTIVE

Our objective is to analyze this data to generate a detailed performance report.:

1. Analyzing the Relationship Between Advertising Budget and Revenue
2. Identifying the Best Advertising Channels
3. Understanding Audience Profiles
4. Analyzing Monthly Campaign & Revenue Trends
5. Comparing Revenue Performance across Dimensions

LOAD THE DATASET

	CampaignID	Channel	AudienceType	AdBudgetK	CTR	ConversionRate	Month	RevenueK
0	1	Instagram Ads	Gen Z	20	4.56	1.16	Jan	112.21
1	1	Instagram Ads	Interest-Based	15	3.96	2.00	Feb	103.92
2	1	Instagram Ads	Interest-Based	21	4.46	2.49	Mar	228.56
3	2	Google Ads	Broad	16	3.98	1.37	Apr	73.69
4	2	Google Ads	Lookalike	23	3.09	2.18	May	111.36

CHECKING THE MISSING VALUE

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 8 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   CampaignID      5000 non-null   int64
 1   Channel         5000 non-null   object
 2   AudienceType    5000 non-null   object
 3   AdBudgetK       5000 non-null   int64
 4   CTR             5000 non-null   float64
 5   ConversionRate  5000 non-null   float64
 6   Month           5000 non-null   object
 7   RevenueK        5000 non-null   float64
dtypes: float64(3), int64(2), object(3)
memory usage: 312.6+ KB
```

- There is no missing value in data table, we don't need any missing data handling process
- Dataset has 8 column and 5000 rows

DESCRIPTIVE STATISTICS

	CampaignID	AdBudgetK	CTR	ConversionRate	RevenueK
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	2.490600	14.042200	5.829684	2.325588	192.097354
std	1.110747	4.639776	2.954297	1.365651	166.033129
min	1.000000	0.000000	0.000000	0.000000	0.000000
25%	2.000000	11.000000	3.790000	1.410000	84.775000
50%	2.000000	14.000000	5.270000	2.150000	148.290000
75%	3.000000	17.000000	7.260000	3.030000	250.192500
max	4.000000	24.000000	21.460000	13.900000	1814.690000

- The average revenue generated from all advertising channels during 2024 is $192.09K
- Several ad channels remained unbudgeted throughout the initial phase of the bootcamp marketing campaign

	CampaignID	Channel	AudienceType	AdBudgetK	CTR	ConversionRate	Month	RevenueK
1343	1	Instagram Ads	Lookalike	21	8.78	13.9	Dec	1814.69

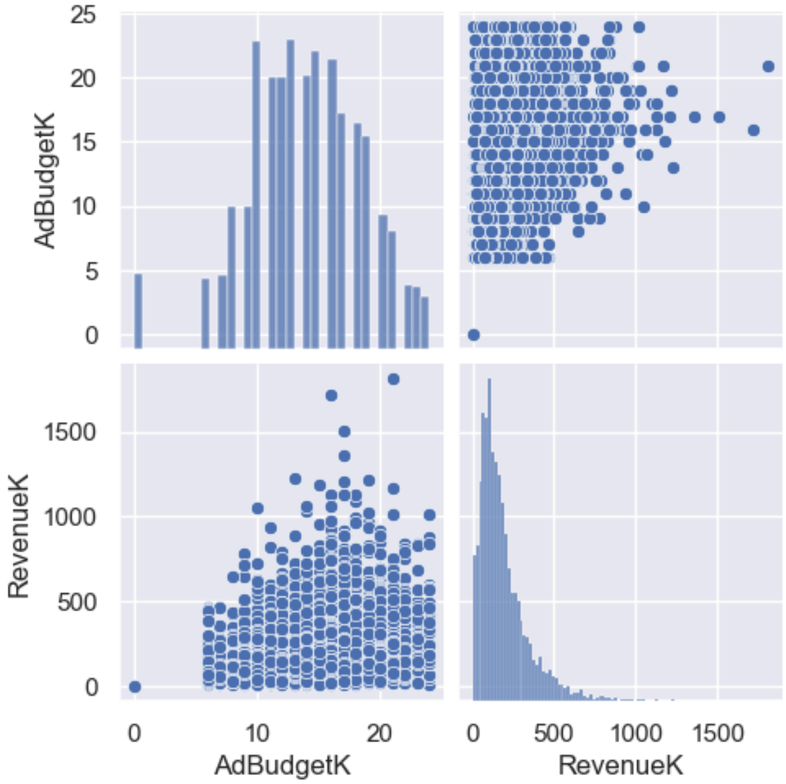
- According to the data, it has the highest revenue through the Instagram Ads channel with a Lookalike Audience of **$1814.69K**

	CampaignID	Channel	AudienceType	AdBudgetK	CTR	ConversionRate	Month	RevenueK
23	4	TikTok Ads	Interest-Based	0	0.0	0.0	Dec	0.0
31	2	Google Ads	Broad	0	0.0	0.0	Aug	0.0
50	4	TikTok Ads	Gen Z	0	0.0	0.0	Mar	0.0
155	3	Facebook Ads	Interest-Based	0	0.0	0.0	Dec	0.0
258	2	Google Ads	Broad	0	0.0	0.0	Jul	0.0
...	...	...	...	...	...	...	...	...
4745	1	Instagram Ads	Lookalike	0	0.0	0.0	Jun	0.0
4759	4	TikTok Ads	Retargeting	0	0.0	0.0	Aug	0.0
4761	2	Google Ads	Interest-Based	0	0.0	0.0	Oct	0.0
4955	3	Facebook Ads	Retargeting	0	0.0	0.0	Dec	0.0
4990	1	Instagram Ads	Gen Z	0	0.0	0.0	Nov	0.0

105 rows × 8 columns

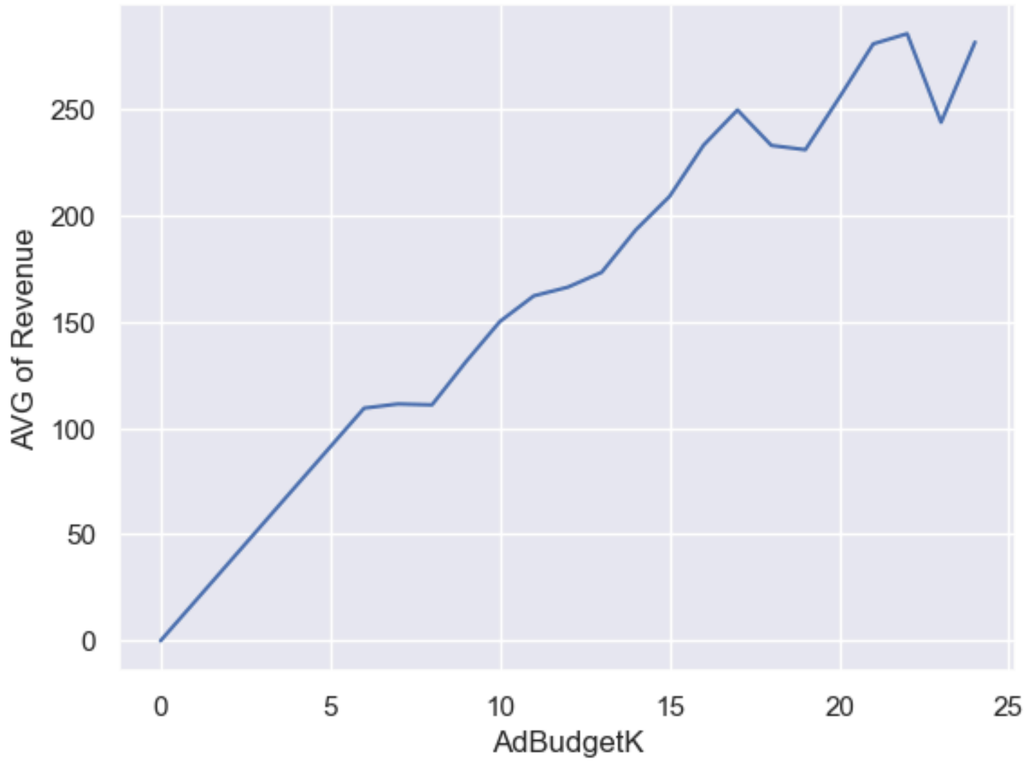
- Approximately 105 channels generated no revenue. Based on that descriptive analysis, this was due to the lack of campaign budget allocation during those months.

3. ANALYZING THE RELATIONSHIP BETWEEN ADVERTISING BUDGET AND REVENUE



	AdBudgetK	RevenueK
AdBudgetK	1.000000	0.321172
RevenueK	0.321172	1.000000

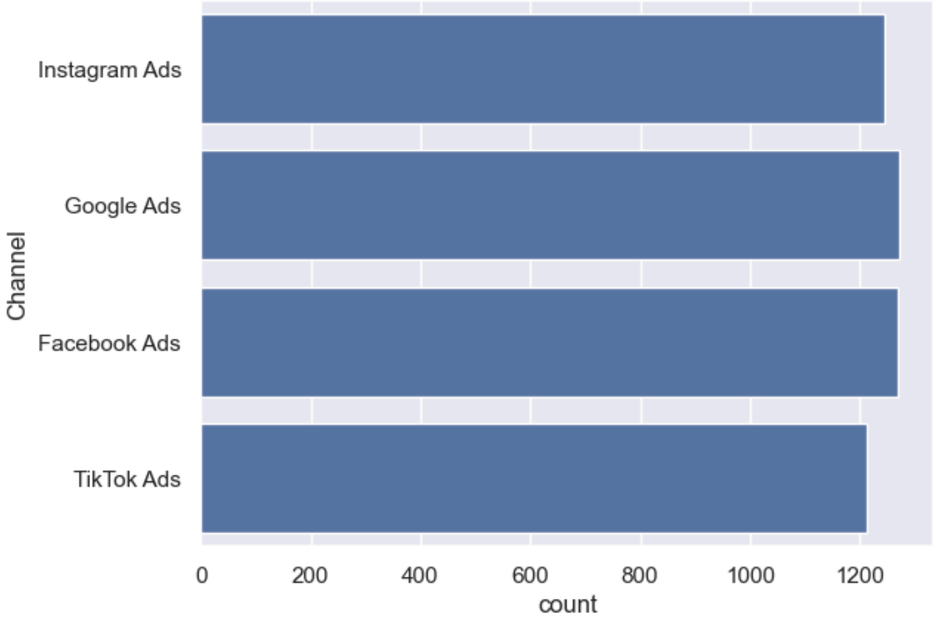
- An increase in AdBudget has the potential to increase income, but the relationship is not linear. There are times when a large AdBudget generates low income.
- The correlation between Adbudget and revenue is 0.32. This is a weak positive correlation. We can say that advertising does contribute positively to revenue, but its contribution is not dominant, as there are other factors that impact it.



- In general, the graph moves to the top right. This means that the larger the allocated AdBudget, the higher the revenue generated. However, there is a decrease in revenue when the budget is increased. This can indicate the efficiency of the ad budget for a particular ads.

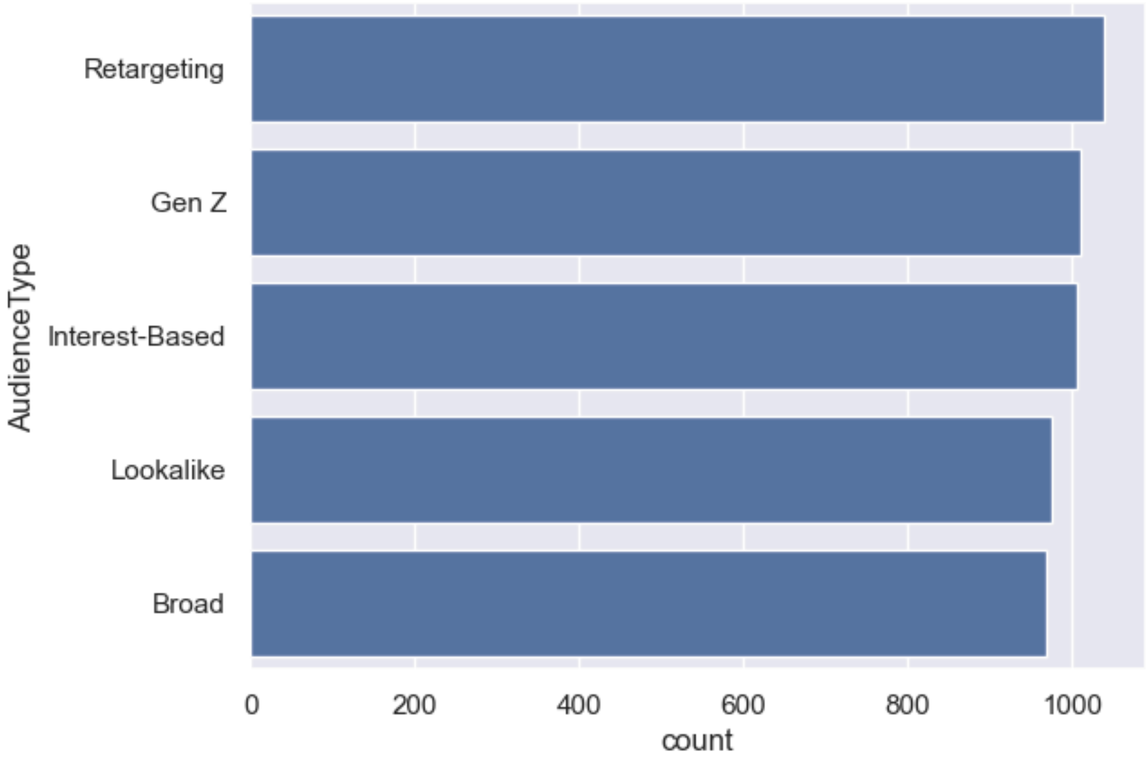
CATEGORICAL VARIABLES

1. CHANNEL



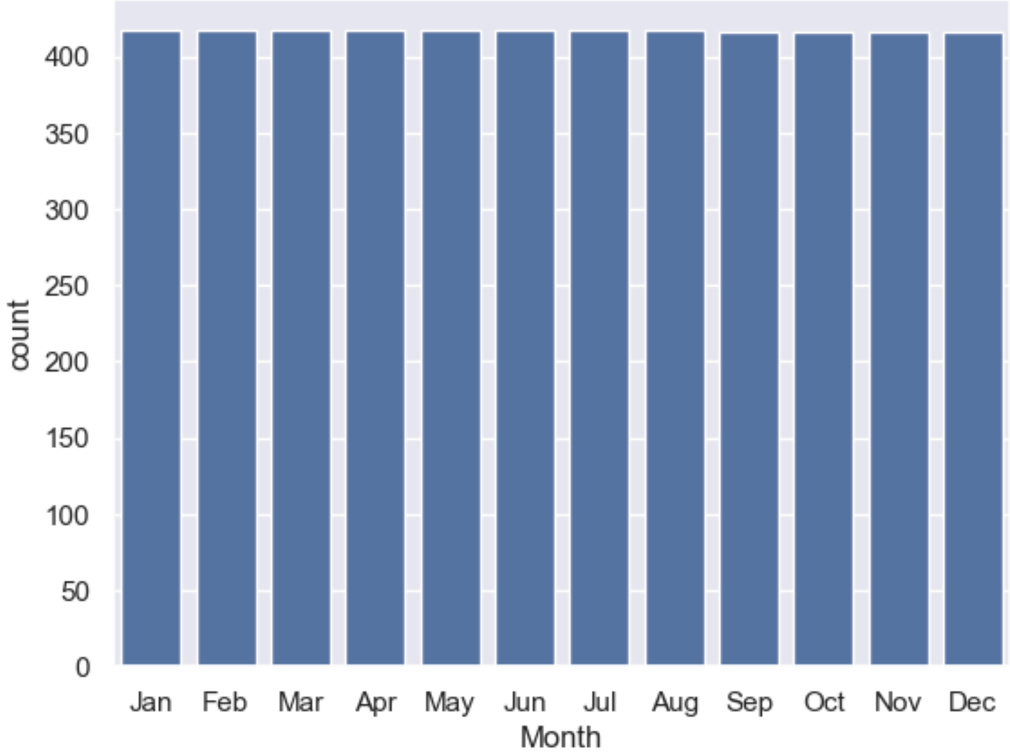
- There is no difference in class sales between the four channels, but TikTokAds is lower than the other three channels.

2. AUDIENCE TYPE



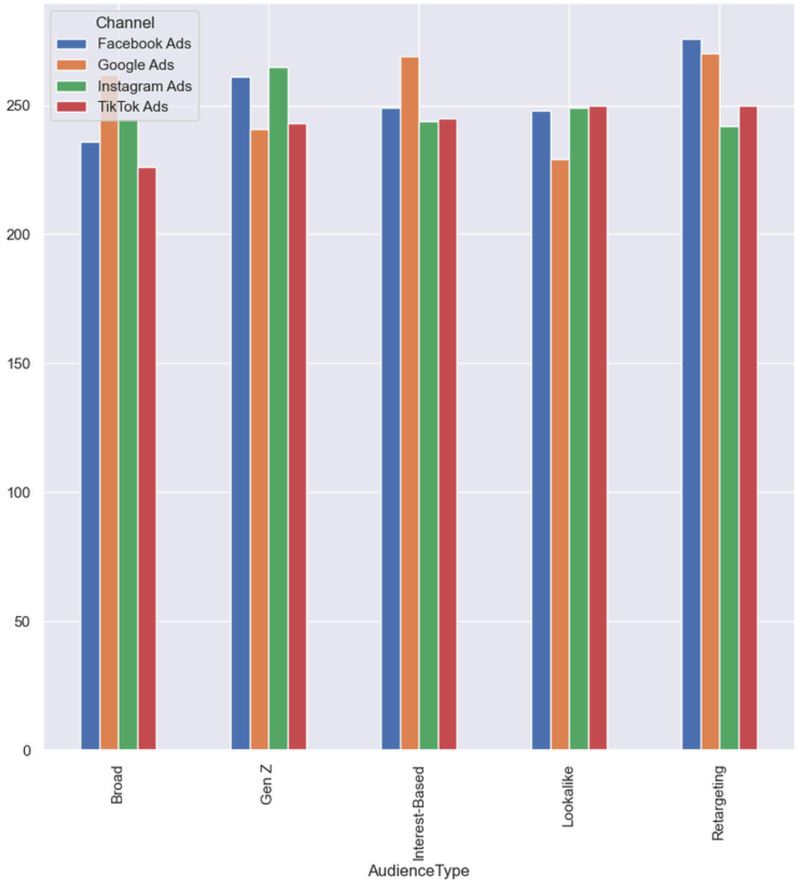
- Based on the identification of audience types, the audience has a balanced distribution across the five main categories, with the largest is **regerting** and the smallest is **broad**.

3. MONTH



- in this campaign there is a balanced sales every month

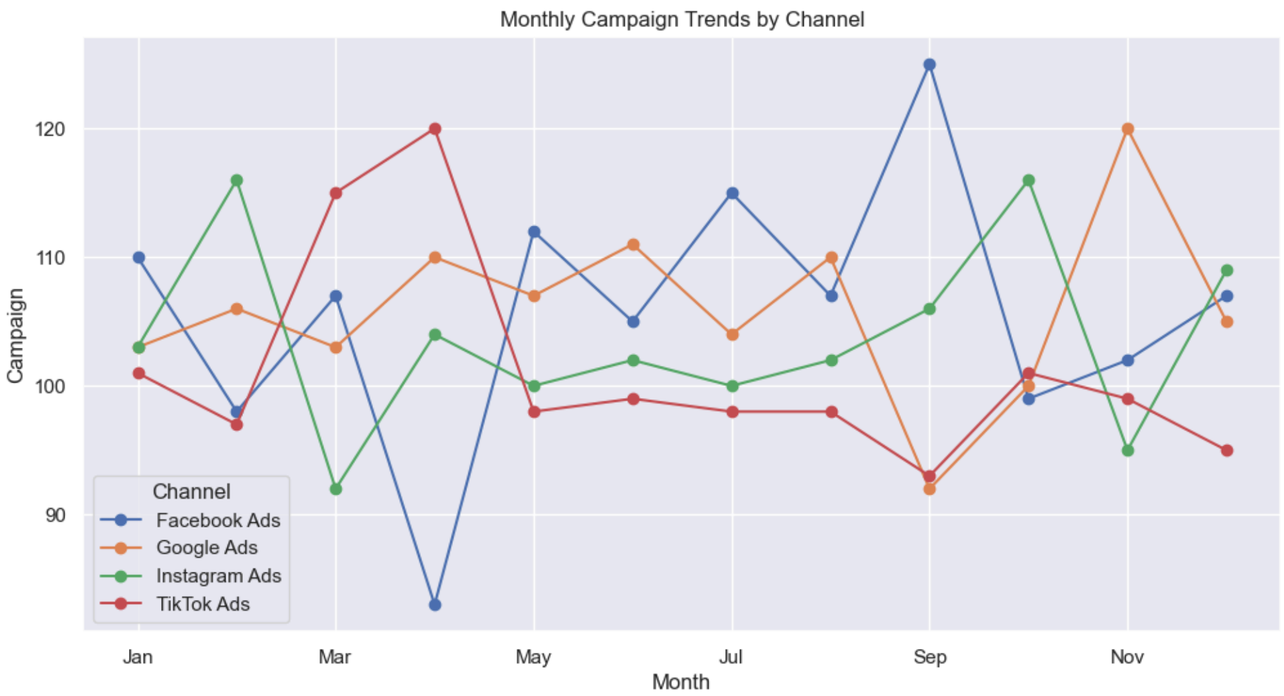
4. AUDIENCE TYPE VS CHANNEL



Channel	Facebook Ads	Google Ads	Instagram Ads	TikTok Ads
AudienceType				
Broad	4.72	5.24	4.90	4.52
Gen Z	5.22	4.82	5.30	4.86
Interest-Based	4.98	5.38	4.88	4.90
Lookalike	4.96	4.58	4.98	5.00
Retargeting	5.52	5.40	4.84	5.00

- Retargeting strategies on Facebook and Google platforms are the highest revenue contributors, while Broad audiences on TikTok require evaluation due to their lowest performance.
- Instagram remains the most consistent top choice for targeting Gen Z.

5. MONTH VS CHANNEL



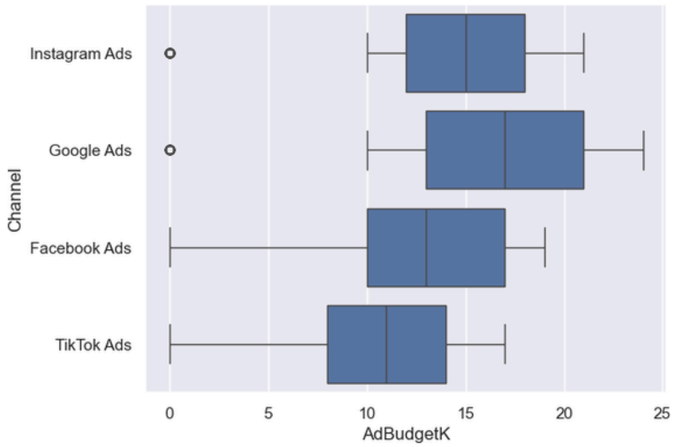
Channel	Facebook Ads	Google Ads	Instagram Ads	TikTok Ads
Month				
Jan	110	103	103	101
Feb	98	106	116	97
Mar	107	103	92	115
Apr	83	110	104	120
May	112	107	100	98
Jun	105	111	102	99
Jul	115	104	100	98
Aug	107	110	102	98
Sep	125	92	106	93
Oct	99	100	116	101
Nov	102	120	95	99
Dec	107	105	109	95

- Facebook Ads showed the most extreme fluctuations. It reached its lowest point in April (83 campaigns), but then surged sharply to its highest in September (125 campaigns).
- TikTok Ads seemed very active at the beginning of the year (March-April), but the intensity continued to decline consistently until the end of the year (reaching 95 in December)

CATEGORICAL AND NUMERICAL VARIABLES

1. CHANNEL VS ADBUDGETK

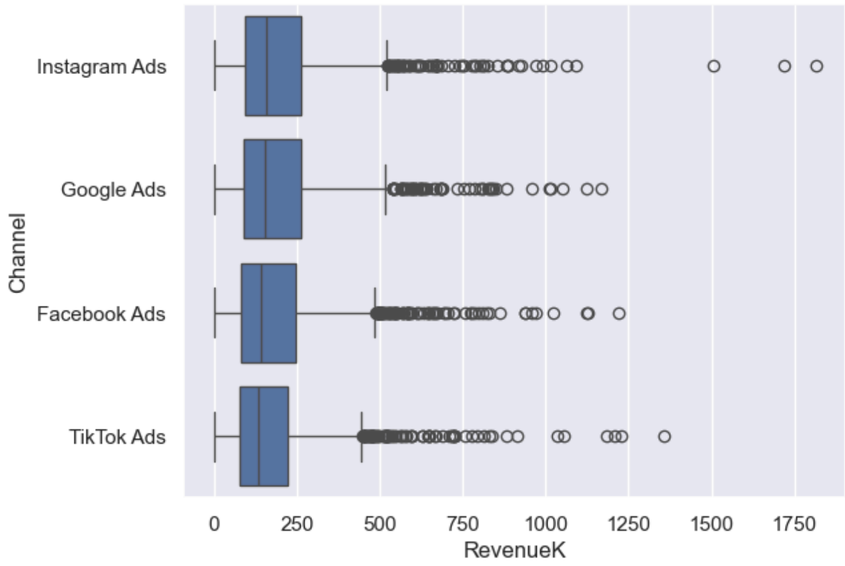
	AdBudgetK			
	count	mean	median	sum
Channel				
Facebook Ads	1270	13.310236	13.0	16904
Google Ads	1271	16.578285	17.0	21071
Instagram Ads	1245	15.038554	15.0	18723
TikTok Ads	1214	11.130972	11.0	13513



- Google Ads has the highest Ad budget for campaign
- TikTok Ads have the lowest budget
- The Instagram Ads and Google Ads channels have no budget allocated. This appears to be the cause of the outliers observed in the boxplot.

2. CHANNEL VS REVENUEK

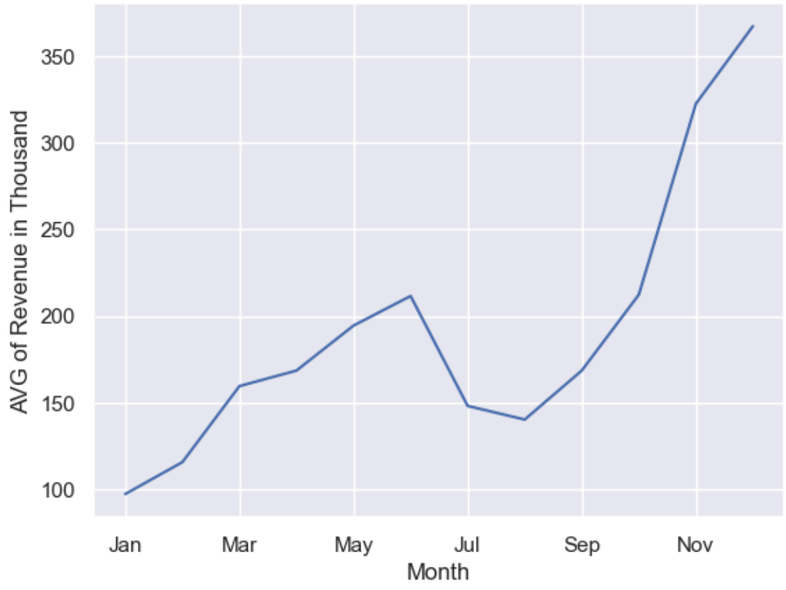
	RevenueK			
	count	mean	median	sum
Channel				
Facebook Ads	1270	187.686803	140.21	238362.24
Google Ads	1271	197.886876	154.25	251514.22
Instagram Ads	1245	204.981807	159.05	255202.35
TikTok Ads	1214	177.436540	134.12	215407.96

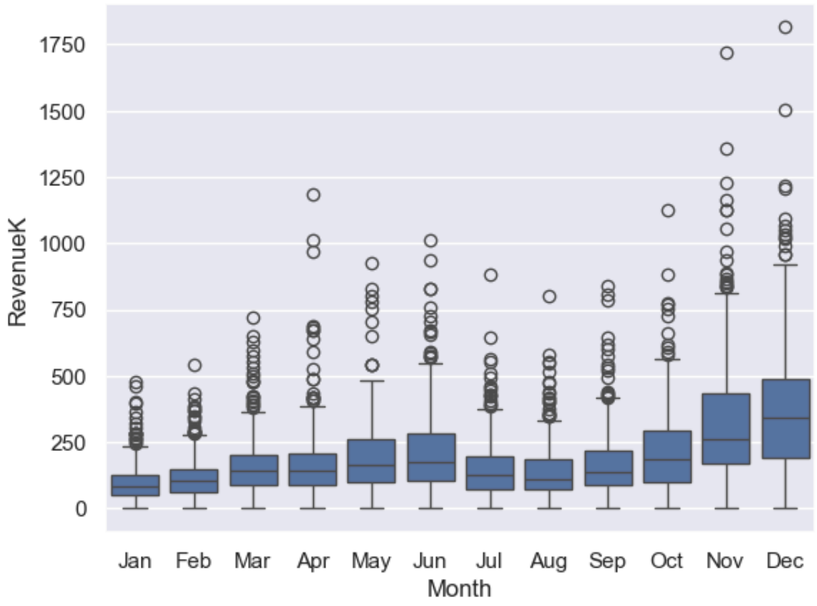


- Instagram Ads generate the highest revenue on average.
- TikTok Ads generate the lowest revenue

3. MONTH VS REVENUEK

	RevenueK			
	count	mean	median	sum
Month				
Jan	417	97.165635	85.60	40518.07
Feb	417	115.590384	103.36	48201.19
Mar	417	159.504484	142.63	66513.37
Apr	417	168.488297	144.35	70259.62
May	417	194.477290	164.59	81097.03
Jun	417	211.556595	177.41	88219.10
Jul	417	148.056571	125.85	61739.59
Aug	417	140.182494	112.54	58456.10
Sep	416	168.480601	138.93	70087.93
Oct	416	212.314760	184.64	88322.94
Nov	416	322.634904	259.51	134216.12
Dec	416	367.441611	341.63	152855.71

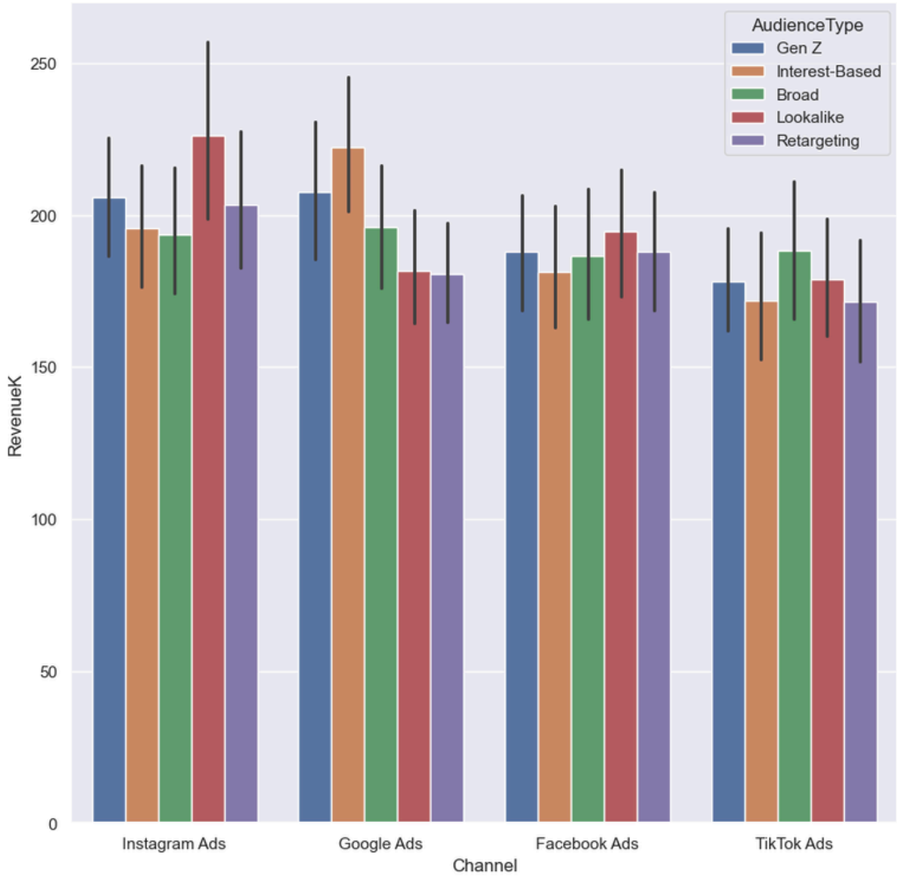


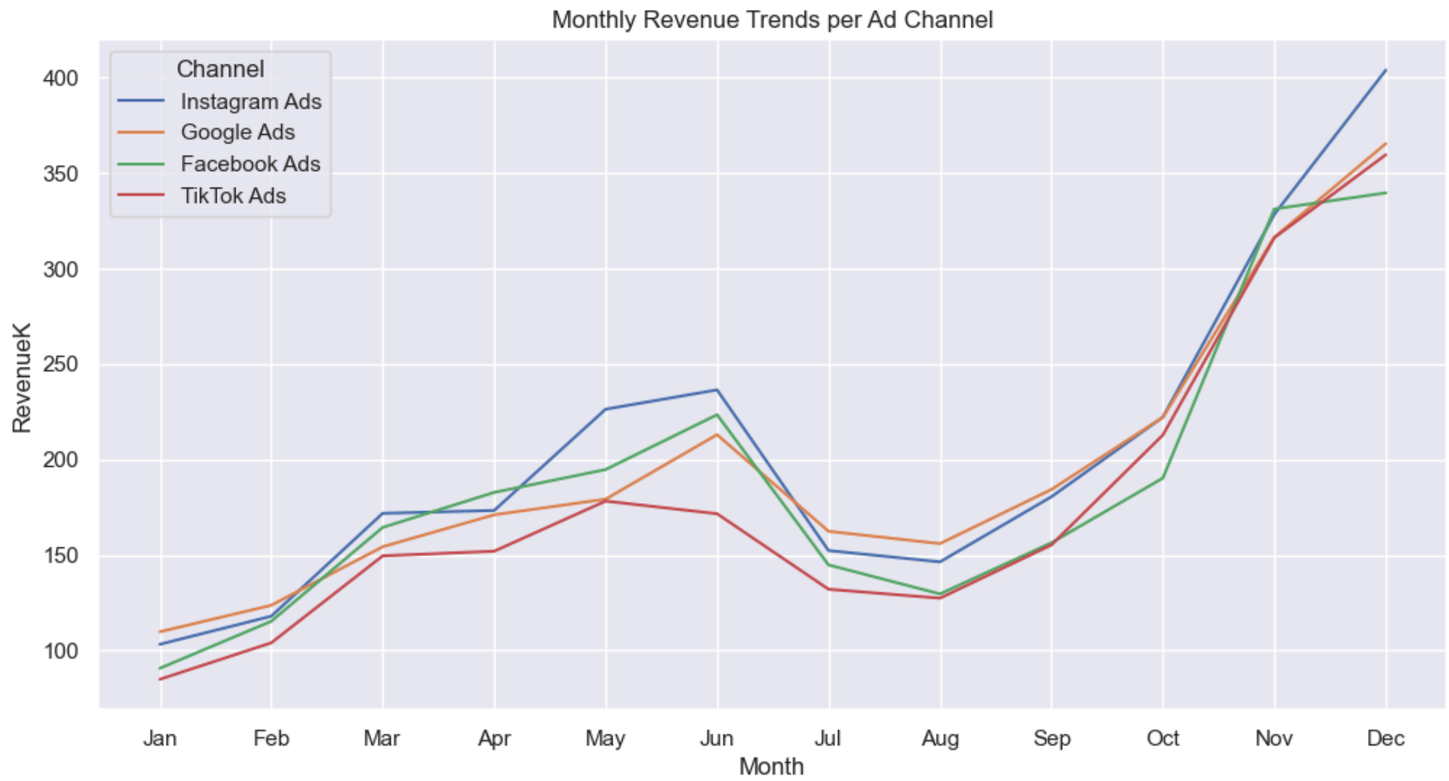


- In general, average revenue shows a significant increasing trend from the beginning of the year to the end of the year.
- There was a significant increase in November and December
- At the end of the year, performance across ad campaigns became increasingly variable, indicating a successful end-of-year promotional effect for certain campaigns.

4. CHANNEL VS REVENUEK VS AUDIENCE TYPE

- Lookalike targeting strategy delivers the highest revenue results on Instagram Ads compared to other platforms.
- For Interest-Based and Gen Z targeting, Google Ads performed the most competitively, making it ideal for capturing audiences with specific purchase intent.





- Instagram Ads showed the most explosive performance at the end of the year
- All advertising channels have a similar trend pattern, namely experiencing a gradual increase from January to June, then experiencing a decline in July and increasing again in the last quarter.

BUSINESS STRATEGY RECOMMENDATIONS

- Reviewing 105 inactive channels. If they have potential, a minimum budget allocation is required to avoid wasted revenue opportunities.
- Given the strong **End-of-Year Effect**, fourth-quarter promotional strategies should be prepared early with bolder budget scalability on Instagram and Google.
- Evaluate TikTok Ads performance thoroughly. If the downward trend continues, consider reallocating the remaining budget to Google Ads (Interest targeting) or Instagram (Lookalike audiences).
- We need to reconsider the efficiency of our spend on TikTok Ads, we can focus on using only three types of channels

THANK YOU FOR YOUR TIME

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DATA ANALYST PRESENTATION